

Coding & Solution

Field	Value
Date	05 July 2026
Team ID	(Your Team ID)
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Field	Details
Repository Link / URL	https://github.com/EjjipurapuKavya/AI-ML-and-GEN-AI-Track-Project-Template-Kavya
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask, Scikit-learn, Pandas, NumPy
Key Features Implemented	• Crop Recommendation• Fertilizer Recommendation• Yield Prediction• Soil & Weather Data Analysis• Farmer Dashboard
Pending / Incomplete Features	• Real-time Weather API Integration• Multi-language Support• Mobile Responsive Improvements
Setup / Run Instructions	1. Install Python dependencies using <code>pip install -r requirements.txt</code>. 2. Run the Flask application using <code>python app.py</code>. 3. Open the application in a browser at http://localhost:5000. 4. Enter farm details to receive recommendations and predictions.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions/classes	Yes

S.No	Criteria	Status (Yes / No)
2	Meaningful variable and function names are used	Yes
3	Code includes comments/documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes <i>(after pushing to GitHub)</i>

Additional Notes / Comments

- The project follows a modular architecture for easy maintenance and future enhancements.
- Machine learning models are integrated to provide accurate crop, fertilizer, and yield recommendations.
- The code is structured for readability, reusability, and scalability.
- Future improvements include cloud deployment, live weather integration, and multilingual support.